A Physical Classroom Whiteboard Environment

Construction, coupled mechanics, observations and reproducible evaluation

Gym Anything for Robotics

Abstract

This environment couples a floating, articulated Unitree G1 with Dex3 hands, six capped markers, a magnetic eraser and a wall-mounted classroom whiteboard. Actions command robot joints; contact determines tool motion and ink transfer. The benchmark evaluates a final coloured surface, not whether a script followed a requested drawing path. Newton 1.5 owns rigid dynamics; body-local tetrahedral elasticity supplies nib, pad and cap-interface reactions, and a conserved pigment state drives both evaluation and camera appearance. A separate renderer never advances simulation time. This dossier states the actual mechanisms, parameter provenance, assisted experimental apparatus, measured results and known boundaries. No autonomous task solution or physical-world calibration is a prerequisite for the environment work; neither is implied by a successful component experiment.

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1 Purpose and acceptance boundary

The deliverable is a realistic digital environment in which a robot policy can attempt classroom tool use. It is not a solved drawing demonstration. A policy must control the robot, obtain physically plausible contact, tolerate occlusion, and leave the requested result on the board. Tools are not attached to hands by hidden constraints. An external inspection camera and an externally supported test hand are useful experimental apparatus, not observations or capabilities of the deployed robot.

Three questions must be kept separate. First, does the digital model implement the intended mechanics without duplicated forces, unbounded state or incorrect causal links? Second, do its dimensions and material choices follow the best available online evidence and defensible estimates? Third, can a particular policy solve a task? This report addresses the first two. A difficult initial view or an unsuccessful policy does not by itself establish an environment bug.

Hardware experiments are outside the project's means. Manufacturer dimensions, official robot assets, primary publications and openly licensed visual references are the evidence base. Where those sources do not determine a value, the report labels a specific engineering estimate and states its operational effect. Demanding unavailable physical calibration would not close a software gap; nor does labelling an estimate excuse missing contact, geometry or data access.

2 Reference scene and provenance

2.1 Classroom, board and support relationships

The room is authored in metres with floor, walls, ceiling, windows, a door, lighting fixtures and movable classroom furniture. The board is wall-supported; its frame and tray have thickness and collision geometry. Marker bodies rest in the tray under gravity. Caps are separate bodies, initially over their own nibs. The eraser has a rigid holder, a compliant pad and an explicitly modelled magnet. Chairs and desks have their own mass, floor contact and finite inertias. No background photograph stands in for physical room geometry.

The board follows the MooreCo porcelain-on-steel family rather than a generic white rectangle[1]. Its nominal outside dimensions are 2.4384×1.2192 m; the bottom is at 0.9144 m above the floor. A 22.225 mm trim reduces the writable dimensions to 2.39395×1.17475 m. The nominal board centre is $z = 1.524$ m, and the writing face is the plane $y = 0.095$ m. The board is a fixed wall-mounted boundary: reaction forces are recorded at that boundary rather than integrated as free-panel motion. Manufacturer tables that describe shipping weight are not used as a measured bare-board mass.

World $+Z$ is up. The classroom extends toward positive Y from the board wall. The board's outward normal is $+Y$; approaching the face means moving toward decreasing Y. Writing coordinates are world (X, Z) expressed relative to the board centre. Model-local marker $+Z$ runs toward the nib, while the eraser's pad faces local $-Z$. These conventions are exposed with geometry so that a teacher does not have to infer the long axis from an image.

2.2 Selected tools and what the product pages establish

The STAEDTLER 351 B reference specifies a capped envelope of $138 \times 17 \times 17$ mm, total mass 19.6 g, polypropylene barrel and cap, and a chisel line-width range of 2–5 mm[2]. The simulation splits that total into a 16.2 g marker and 3.4 g cap; the split is an estimate, not a separately published mass measurement. A 14 mm barrel, coloured tail and collar, finite nib and open cap shell account for the envelope. The nib is $5 \times 2 \times 7.5$ mm with its bonded root at local $z = 55.5$ mm.

The 652 wiper reference supplies a $107 \times 57 \times 20$ mm envelope and a magnetic inlay[3]. The holder and $103 \times 53 \times 5$ mm pad are separate mechanical regions. Its 30 g total mass and magnetic force law are declared estimates. The product dimensions do not determine pad modulus, magnetic gap dependence or friction. Those values therefore appear in the parameter ledger below rather than being attributed to the manufacturer.

The selected marker family has six colours: black, red, blue, green, orange and violet. Digital colour triplets are appearance choices, not spectrophotometry. They identify separate pigment inventories and scoring channels;

changing colours requires using the corresponding physical marker. No policy endpoint changes an already-held marker's pigment identity.

2.3 Robot and camera

The pinned G1 asset has 29 actuated body joints and fourteen actuated hand joints. Each Dex3 has a three-joint thumb and two joints each for index and middle fingers. The 43-coordinate actuator inventory is checked at import. The base remains a floating rigid body. Floor support comes from feet and contact, not a base weld. The robot's native mesh and inertia definitions are retained with the asset's joint limits[4, 5].

The advertised G1 EDU maximum knee torque and the pinned URDF knee limit differ: the former is 120 N m and the latter 139 N m. The implemented limit is the pinned asset's value. This discrepancy is reported, not silently reconciled by altering the actuator. The marketed hand sensor options are not automatically present in policy observations. This benchmark does not claim fitted tactile arrays merely because the hand product can be ordered with them.

The onboard RGB frame is attached through the URDF D435 mount. Its nominal optical reference is the D435 RGB camera, with 69-degree horizontal field of view and a 1920-by-1080 reference stream[6]. Export and live-render sizes are declared separately. A 960-by-540 frame is a lower-resolution observation, not a new lens or a displaced camera. The present image formation is pinhole RGB; rolling-shutter timing, sensor noise and stereo-depth failure are not fully simulated. No perfect depth or external classroom camera is silently supplied to the restricted policy.

2.4 Parameter ledger

Table 1 separates four provenance classes: M (manufacturer), A (revision-pinned asset), D (derived from explicit geometry), and E (engineering estimate). An E value is usable within this digital benchmark; it is not a measurement of the named object pair. Fixed-point, refinement and energy checks test the implementation around these values, not their laboratory accuracy.

Table 1: Principal parameters and provenance. SI unless shown otherwise.

Quantity	Implemented value	Provenance and role
Board outside dimensions	2.4384×1.2192 m	M, MooreCo 8-by-4-foot family.
Board bottom / face	0.9144 m / $y = 0.095$ m	E placement using classroom mounting guidance.
Trim / backer	22.225 / 11.1125 mm	M reference construction, explicit digital section.
Capped marker mass	19.6 g	M, STAEDTLER individual marker, not multipack.
Marker / cap mass split	16.2 / 3.4 g	E split preserving the specified sum.
Barrel radius	7 mm	E internal profile inside the specified outer envelope.
Nib geometry	$5 \times 2 \times 7.5$ mm	E explicit chisel specimen, not product tomography.
Nib Young's modulus / Poisson	0.9 MPa / 0.10	E compliant porous-nib approximation.
Pad Young's modulus / Poisson	2.2 kPa / 0.10	E soft felt approximation.
Cap Young's modulus / Poisson	1.3 GPa / 0.42	E generic polypropylene, not selected-cap test.
Nib / pad friction	0.35 / 0.55	E contact-pair estimates; not copied as measured from unrelated pens.
Nib / pad normal damping	0.45 / 1.0 N s/m	E dissipative stabilization of local contact.
Finger / tool friction	0.45	E ordinary collision coefficient, unchanged for hold tests.

Quantity	Implemented value	Provenance and role
Eraser dimensions	$107 \times 57 \times 20$ mm	M, STAEDTLER 652.
Eraser / chair / desk mass	0.030 / 4.7 / 12 kg	E, explicitly separated from dimensions.
Magnet surface force / gap scale	1.4 N / 1.5 mm	E finite-gap attraction.
Ink inventory per marker	1500 μl	E finite transferable pigment-equivalent inventory.
Equivalent full-film thickness	3 μm	E conversion between coverage and inventory.
Drying time / erasure distance	12 s / 18 mm	E transfer kinetics; not a cap-off lifetime.
G1 joint, mesh, inertia data	43 actuated joints	A, SHA-pinned official URDF and meshes.
Writing texel spacing	0.5845 mm horizontally	D, writable 2.39395/4096; 4-by-4 material subsamples.
Physics timestep	0.5 ms	Numerical choice checked against 0.25 ms.

2.5 Research that informed the design

Multi-pen robotic drawing demonstrates why marker pose cannot be assumed rigidly known after pickup: grasp uncertainty, exchange and feedback matter[7]. Impedance-controlled drawing emphasizes compliant normal contact alongside tangential motion, but does not supply a universal pressure-to-width calibration for every tool[8]. The Flying Calligrapher's custom pen is useful as an example of separate load, friction and stroke experiments, not as a source of STAEDTLER-on-porcelain constants[9]. Consequently this environment measures normal load, slip, contact area, transfer and release separately, then combines them through an articulated hand. It does not borrow a successful controller trajectory as a substitute for the mechanics.

3 Mechanical model and ownership

3.1 One stepping authority

Newton 1.5 builds the articulated scene and supplies the rigid solver interface. The default CPU backend is SolverMuJoCo using MuJoCo 3.11; the explicit GPU mode uses MuJoCo Warp. Isaac Sim 5.1 is used in a separate process solely for rendering. PhysX does not integrate this environment. Changing the renderer's update count does not create a robot action or advance Newton's clock.

At each physics tick the environment validates the requested action, evaluates its custom tool reactions, integrates the rigid world exactly once, transfers ink using that contact interval, checks the resulting finite state, and records the post-step history. Native finger, furniture and floor contacts remain in the rigid solver. Nib–board, pad–board and cap-interior contacts have explicitly assigned constitutive owners. Collision masks remove only the duplicate native response for those owned interfaces; external cap impacts and ordinary tool collisions still use native contact.

For generalized velocity v, the rigid equations have the usual structure

$$M(q)\dot{v} + C(q, v) + g(q) = \tau_{\text{act}} + J_n^T f_n + J_t^T f_t + J_b^T f_{\text{tool}}. \tag{1}$$

The imported motor limits bound the joint servo outputs. The default position gains are 800 N m/rad for leg joints, 120 for the remaining arm joints, 80 for wrists and 1.8 for finger joints; corresponding damping is 20, 6, 3 and 0.035 N m s/rad. These are controller choices, distinct from URDF effort limits. A target is not a teleport: actual joint motion depends on gravity, inertia, contact and torque saturation. IK computes targets using private preview data; executing them still goes through physical servos.

3.2 Resolved local elasticity

Nib and felt use linear tetrahedral displacement elements (P1); the thin cap shell uses quadratic elements (P2). The solids are expressed in tool-local coordinates and have bonded root nodes. Their mass is already included

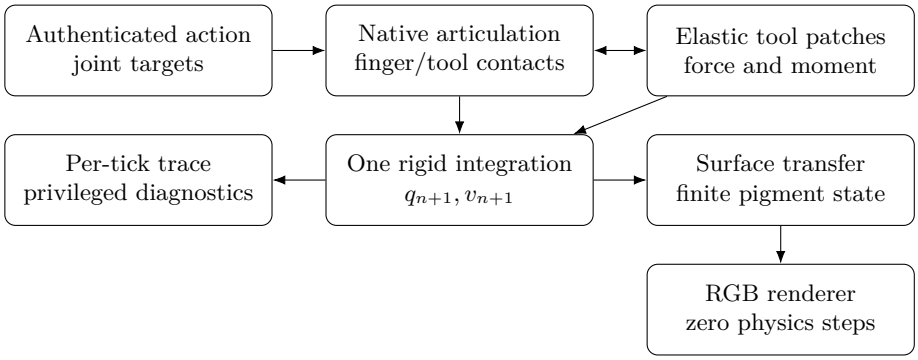


Figure 1: Ownership, not separate sequential robot motions. Constitutive reactions enter the same rigid step; rendering consumes state without stepping.

in the authored rigid tool inertia. There is no additional separately integrated soft body whose mass would be counted twice. Internal deformation equilibrates quasi-statically within the rigid step.

With displacement field u, small strain is $\epsilon = \frac{1}{2}(\nabla u + \nabla u^T)$ and isotropic stress is $\sigma = 2\mu\epsilon + \lambda \operatorname{tr}(\epsilon) I$, where

$$\mu = \frac{E}{2(1+\nu)}, \qquad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad K = \sum_e \int_{V_e} B_e^T D B_e \, dV. \tag{2}$$

Dirichlet elimination removes bonded degrees of freedom exactly. The resulting free stiffness matrix is factorized once. Geometry-dependent contact operators are cached within explicit memory budgets, but deformed states are not cached as if they were current. P2 integration uses four tetrahedral quadrature points.

For candidate contact nodes, projected normal displacement must remove their penetration. Condensing the elastic degrees of freedom gives a compliance matrix C. Normal loads solve the convex unilateral problem

$$\min_{\ell \geq 0} \ \tfrac{1}{2}\ell^T C \ell - p^T \ell, \quad g = C\ell - p \geq 0, \quad \ell_i g_i = 0. \tag{3}$$

The active-set method checks previously inactive nodes as well as loaded nodes. It enforces a 2×10^{-9} m contact residual tolerance and rejects a failed solve. The condensed and reconstructed deformation must agree. Elastic energy is $U_e = \frac{1}{2} u^T K u$. This supplies an observable independent of the final rendered stroke or whether the fingers retained a tool.

The implementation guards compression above 60% of the nominal nib or pad depth and maximum component strain above 0.60. These are rejection boundaries, not a claim that linear elasticity is accurate to 60% strain. Below the guard, small-strain modelling still becomes less representative as deformation grows. Crushing, tearing, plastic failure and free soft-body vibration are not included. Actual peak strain and refinement results therefore matter alongside the guard.

3.3 Friction, load coupling and energy

For tangential slip velocity v_t, regularization speed $v_\epsilon = 1$ mm/s and node normal load ℓ, the board-friction law is

$$f_t = -\mu \ell \frac{v_t}{\sqrt{\|v_t\|^2 + v_\epsilon^2}}. \tag{4}$$

It permits slow creep rather than ideal sticking. The mass-scaled implicit velocity solve minimizes a strictly convex sum of kinetic-distance and friction potential, with a 10^{-12} scaled momentum-gradient tolerance. Cap friction uses a separate bounded impulse sweep and is not assigned this board law.

Node loads are combined with dissipative normal damping, distributed using the elastic load weights. Tangential response is solved implicitly with a smooth regularized Coulomb magnitude rather than an abrupt sign change. Normal and tangential deformation are coupled by a fixed-point iteration; the implementation requires force change below 10^{-7} N within 30 iterations. Nonconvergence is an explicit failed physics state, not permission to drop friction or retry the action with a different coefficient.

For forces f_i at world points r_i and tool centre of mass r_c, the rigid tool receives

$$F = \sum_i f_i, \qquad T = \sum_i (r_i - r_c) \times f_i. \tag{5}$$

The immovable board receives the equal-and-opposite reaction, expressed about the world origin in the audit. This separates an external wall reaction from a missing conservation term. A hand-held marker is affected through its actual finger contact, so board load can cause tool slip and change robot joint effort.

The energy ledger includes translational and rotational kinetic energy, gravitational potential, elastic storage, actuator work, custom contact work and boundary impulse terms. A residual is reported numerically rather than declared zero by construction. Tests distinguish dissipative friction from energy injection, external fixture work from actuation and fixed-wall reaction from free-body momentum. Native contact impulses are read from the solver; they are not reconstructed from visual overlap.

3.4 Cap retention and eraser attraction

The cap's inner bore has three explicit retention ribs. In the reference mesh, the nominal inner radius is 7.55 mm with up to 0.60 mm inward rib depth, and the outer radius is 8.5 mm. The default P2 discretization has 72 circumferential, 12 axial and two radial subdivisions. These rib details are an engineering representation of a snap-fit shell, not a scan of a commercial cap.

Cap–barrel reaction is generated from this elastic interference and friction. There is no binary "attached" joint removed at a chosen separation force. As overlap and rib contact end, the reaction naturally vanishes and the cap continues as a free rigid body. A pull-off experiment must record the applied load, relative axial position, remaining contact and marker motion; a cap moving in a render alone cannot distinguish elastic release from a slipping marker or a fixture that moved both parts.

The eraser magnet produces finite attraction near the steel-backed board, not a permanent weld. Gravity and pad friction determine whether it stays in place or slides. The pad is deformable against the board, while its other external contacts retain the documented rigid collision proxy. Demonstrating magnetic retention does not demonstrate hand-held erasing; the latter is a separate combined experiment with the holder transmitted through articulated fingers.

3.5 Exact computational reuse

The elastic stiffness K is fixed for a given material mesh. Its condensed three-by-three response blocks can therefore be reused when contact positions select the same reference nodes. A bounded 16 MiB packed-block cache removes repeated array construction. Each call still projects those blocks using the current normals, solves for the current loads and checks every inactive gap. It does not cache a cap pose, contact force or successful contact decision. The packed cache is invalidated if the candidate-node indexing changes.

Three-vector contraction and left-to-right force accumulation preserve the scalar reference's arithmetic order. Direct contraction comparisons and same-state whole-scene continuations test this reuse separately from performance measurements. A faster result is not accepted merely because the final picture looks similar. Likewise, the public target image may be cached, but current pigment and evaluator scores are always recomputed.

4 Contact-driven surface state

4.1 Persistent pigment and finite inventory

Ink lives in board coordinates, not in a selected camera image. Sparse 128-by-128 tiles hold six pigment channels and wetness timestamps. Each display texel contains sixteen persistent material samples on a 4-by-4 grid. At the default 4096-by-2048 resolution, the horizontal material-sample spacing is approximately 0.146 mm. Sparse allocation makes a few written strokes cheaper than allocating every material field across the board in advance.

The footprint comes from the loaded, oriented nib or pad projected onto the board. An elliptical coordinate mask tapers the boundary; it is clipped to the writable face and excludes trim. Contact continuity determines swept

transfer: the model must not draw a line across a lifted interval simply because a tool was at two locations on successive API calls. The scripted specimen-preparation path used by component tests is inaccessible to the policy.

Let c be a sample's coverage in the active colour, $m \in [0, 1]$ its footprint weight, F normal load in newtons and Δt elapsed contact time. The implemented deposition increment is

$$\Delta c = \max(0, 1 - c)\,[1 - \exp\{-350\Delta t \min(F, 1)m\}]\,. \tag{6}$$

The transfer constant has the corresponding inverse-second/force normalization implicit in this SI implementation. This saturating law prevents a stationary nib from producing unlimited coverage. It is an engineering kinetic model, not fluid flow through individual felt pores. Zero load, zero duration, exhausted inventory or an ineligible contact transfers no pigment.

One fully covered material sample represents

$$V_s = \frac{W_b H_b}{N_x N_y\, 16}\; h_f\, 10^9 \quad [\mu\mathrm{l}], \tag{7}$$

where W_b, H_b are writable dimensions and $h_f = 3 \times 10^{-6}$ m is the equivalent full-film thickness. The sum of requested increments is scaled down if necessary to fit the remaining reservoir. Deposited volume is subtracted from that marker's finite inventory. This is pigment-equivalent bookkeeping; it does not update rigid-body mass as microlitres of material leave the tool.

4.2 Drying and wiping

Each sample records the latest deposition time t_s. Its model wetness is $w = \exp[-\max(0, t - t_s)/12]$. This describes the deposited film. It is explicitly not the marker's cap-off dry-out clock; the selected manufacturer's cap-off storage claims have not been converted into a made-up shared timeout.

For pad travel s, load F, retained pad pigment L in microlitres and the same footprint weight m, the removal fraction is

$$a = \frac{s}{0.018} \min(F/2, 2) \frac{1}{1 + L/120}, \tag{8}$$

$$r = 1 - \exp[-a(1 + 0.5w)m]. \tag{9}$$

Every colour loses fraction r from the contacted samples. Exactly that pigment-equivalent volume is added to the eraser's colour-resolved load. The represented pad becomes less effective as it fills. A hovering pad cannot erase, and stationary contact with zero travel cannot act as a delete button. Pad redeposition, permanent staining and detailed solvent transport are omitted; their absence is not disguised by clearing pixels in a rectangle.

The relevant material conservation check is initial marker inventory equals remaining inventory plus board pigment plus pad pickup, within numerical precision and any explicitly seeded specimen. A test that initializes red pigment directly must record that amount separately. It tests wiping mechanics, not whether a hand previously wrote the patch.

4.3 Appearance and scoring share the same material

For colour-channel density c_j and RGB reference p_j, the sample colour is

$$I = 0.965 \exp\left[-\sum_j c_j\{-\log(\max(p_j, 0.012))\}\right]. \tag{10}$$

The renderer averages sixteen sample colours, then quantizes to an 8-bit RGB texture. This absorption-style approximation darkens overdraw and keeps colour inventories distinct. It is not a complete layered paint model: optical-density addition is commutative, so dry stroke order does not produce opaque top-layer occlusion. That limitation matters when interpreting dark-over-light results. It does not authorize evaluating a different image from the one rendered.

The board texture is uploaded to the live RTX material only when its content changes. Current body transforms and elastic vertices are sent separately. The renderer associates each returned frame with the request sequence and simulation tick. It rejects stale frame identity or any render-induced tick advance. A snapshot restore must also restore the pigment field and invalidate the material image; an old GPU texture would be a correctness bug even if the physics checkpoint were exact.

5 Task definition and information boundary

5.1 Tasks and verifier

Text tasks use a deterministic reference layout, not open-ended OCR. The bridge family uses a multicolour schematic with a declared target image. Reference stroke width is 3 mm. The target is public instruction; current pigment arrays, object state and contact forces are privileged observations.

For each colour, actual material coverage above 0.15 is compared with the colour's reference mask using a 2 mm neighbourhood tolerance. Precision measures how much drawn material is near that reference; recall measures how much of the reference is covered. The per-colour score is

$$F_1 = \frac{2PR}{P+R}, \qquad \text{success} = \bigwedge_{\text{present colours}} (F_1 \geq 0.85). \tag{11}$$

The reported scalar score averages colour scores, but success requires every evaluated colour to pass. An unwanted extra colour is not ignored. Empty output, wrong colour, shifted output and a broadly painted board are explicit negative controls. These controls test the actual material verifier, not a filename or an actor's claim of success.

This metric rewards spatial imitation of a declared target. It does not judge arbitrary artistic merit, handwriting intelligibility across styles or semantic equivalence of a new bridge drawing. The benchmark can test visually guided robotics without pretending to solve those separate assessment problems.

5.2 Actions, reset and clock

The GAR adapter, Gymnasium wrapper, HTTP service and manual dashboard share the same environment object and stepping semantics. Actions are named joint targets or bounded Cartesian requests converted to joint targets. Dex3 is not a parallel two-finger gripper; finger joints are commanded individually. The documented TCP is 80 mm along the wrist-yaw link's local $+X$ axis, not automatically the centre of every object that the hand happens to hold.

Kinematic previews use a separate MuJoCo data object and do not advance the live environment. Their sampled joint path is a check, not a mathematical guarantee of collision-free dynamics. Optional runtime guards inspect execution. A failed preview executes zero ticks. Invalid keys, nonfinite targets, out-of-limit joint requests and a duration not exactly divisible by the timestep are rejected; the API does not silently clamp or round requested values.

The clock advances only through explicit stepping operations, including an explicit no-new-command interval. Observations do not step. Reset creates the declared initial state; restoring a checkpoint is a separate operation with a new continuity event. Neither operation can be used to claim a continuous video across the discontinuity. Continuous component footage records one uninterrupted process and states its sampling stride and playback speed.

5.3 Teacher and restricted policy

Two different bearer keys select roles. The privileged teacher can inspect geometry, object poses, forces, detailed state, history and checkpoints. The restricted policy receives the task instruction, robot proprioception and the mounted camera observations, along with its action interface. It cannot obtain current object transforms, exact meshes, ink fields or review cameras through a help route, an error payload or a less obvious observation route. Test-only object placement and phantom loads are not public capabilities.

The self-description should be treated as the authoritative route and schema listing for the running version. Static prose cannot guarantee that a deployed process has reloaded a newer implementation. Release checks therefore query the running service with both roles, attempt forbidden accesses and exercise real image updates, rather than merely importing the route functions.

5.4 Durable command and substep evidence

History records each executed physics tick, including poses, velocities, joint targets, actuator effort, native contact points and wrenches, custom tool contacts, deformation, pigment metadata and the audit ledger. Command IDs and timestamps connect those frames to before/after action records. This permits a teacher to distinguish a slipping grasp from an earlier cap or board contact. Recorded contact wrench belongs to that integration interval; the pose is its post-step result. They must not be misread as simultaneous static equilibrium.

An optional SQLite store uses WAL and FULL synchronous commits. A 64 MiB memory cache and a default 1 GiB persistent payload budget bound retained evidence. Evictions, failed recording and unfinished commands are explicit. A disk-reserve preflight occurs before mutation; a failure after some executed ticks reports that execution rather than presenting the request as an untouched transaction. A restarted process does not silently replay an interrupted command. Checkpoint recovery is owner-controlled and history records the break in continuity.

6 Experimental design: components before policies

6.1 Why assisted fixtures are the right evidence here

An autonomous robot policy would conflate perception, planning, grasp selection, motor control and mechanics. A failed drawing could then expose any of them. The environment's acceptance instead uses isolated material specimens and labelled combined apparatus. A fixture may initialize a tool in the fingers, support a palm, apply a measured pull or seed a pigment patch. Those operations are recorded, remain outside the policy interface and are not task successes.

There are three levels of mechanical evidence. Material tests check force, deformation and numerical refinement of the constitutive elements. Isolated articulated-hand tests check transmission across unwelded finger/tool contact. Complete-robot tests add the real arm chain, floating body and its reaction path. A successful test at one level is not substituted for another. In particular, a spring load at a free wrist is not evidence that a marker's nib actually contacted the board through that hand.

6.2 Combined experiments and declared apparatus

Experiment	Setup and imposed load	Required physical observations
Loaded hold/release	Fixed test palm; articulated fingers; initial marker placement; axial and lateral nib load.	Retention under the declared load; relative slip; actual free drop when fingers open.
Hand-held writing	Floating isolated hand; recorded six-axis palm spring; marker support only for initial 0.5 s.	Native finger contact, board load, deposited pigment, bounded slip, subsequent withdrawal.
Hand-held cap pull	Same physical grasp; cap pull gauge ramp; marker support removed before pull.	Relative cap release, applied force at release, retained marker, no attachment constraint.
Hand-held erasing	Articulated fingers on holder; known seeded pigment; supported palm approaches, wipes and withdraws.	Pad/board load, pickup into eraser inventory, hand contact and holder slip.
Complete-robot writing contact	Legal initial arm posture selected offline; pelvis suspension and wrist-motion apparatus; no marker support after setup.	Actual finger/tool/board chain, joint effort and full-body state; no autonomous walking or grasp claim.

The palm support is a translational spring and damper plus an orientation spring. Its applied world wrench is sampled alongside the tool pose. Gravity compensation at the apparatus is explicitly external work. In the cap experiment, the gauge increases pull at 20 N/s up to 4 N and is removed after release. The report states whether the marker remained held; otherwise a separated cap could simply reflect an uncontrolled falling marker.

Combined runs retain their initial and intermediate exports and sampled motion. The 20-tick retained component-review cadence corresponds to 10 ms at the default timestep. These component sample files are not an unlimited every-tick archive. The separately tested HTTP recording service retains finer, every-tick history within its explicit storage budget. Sampled component plots alone must not be used to infer an unrecorded event ordering between frames. Gate counters and peak quantities are evaluated every tick where indicated in the raw record.

6.3 Independent numerical checks

Temporal comparisons run the same declared apparatus at 0.5 and 0.25 ms, compare the corresponding simulated times and measure material transfer as well as rigid motion. Spatial comparisons refine the board representation separately from the tool elasticity mesh. Refining both at once would hide which discretization produced a discrepancy. The general relative-observable gate is 2%, with component-specific quantities and reference denominators recorded.

The writing sweep varies normal load, nib angle and speed. Hover, capped, exhausted-supply, erase and overdraw controls prevent a convincing picture from masking an invalid transfer path. Material force and deformation plots use physical quantities, not unit-test assertion counts. Rigid conservation checks include external boundary reactions; full robot tests inspect effort limits, body tilt, joint state and finite contacts.

The default cap mesh is compared with a finer independent discretization. The reported difference refers to a specific insertion-load observable; it is not a bound on every cap force under all orientations. Likewise a stable passive classroom does not establish that every possible high-speed policy collision falls inside the material model's strain domain.

6.4 Performance measurements

Define latency ratio $R = T_{\text{wall}}/T_{\text{simulated}}$. A value of 80 means one simulated second takes eighty wall seconds; it does not mean real time. The requested operational target is $R < 100$. Startup is reported separately. A short passive microbenchmark is useful for diagnosing overhead but cannot establish the full action/camera/recording loop's ratio.

The serial operating experiment uses the full default classroom and robot, 4096-by-2048 material state, durable every-tick recording, real loopback HTTP actions and a persistent 960-by-540 onboard RTX camera with 16 samples. Four 1000-tick actions contribute two simulated seconds. Each cycle includes the action and following observation, and records their separate wall times. No physics settings or history resolution are disabled to improve the number.

The CPU throughput experiment launches exactly four complete worlds, synchronizes their stepping start after construction and advances each for 1000 ticks with durable recording. Aggregate ratio is the slowest worker's timed duration divided by the sum of simulated seconds across the four worlds. This is distinct from per-world latency and from a single GPU-native batch. The GPU experiment also reports its explicit solver settings; GPU support alone is not a speedup claim. Timing runs are isolated from other owned heavy work. The complete-G1 contact fixture separately partitions stepping time into ticks with and without measured nib load. That ratio includes normal per-tick in-memory history, but excludes fixture readback and USD export. It is not substituted for the HTTP/RGB/durable-history ratio.

6.5 Failures, provenance and statistical scope

The campaign records its effective configuration and raw measurements. Failed initial placements and failed solves are retained. A fixture change creates another identified experiment; it does not overwrite the unsuccessful one. An export error before the first physics step is a harness failure, distinct from a numerical failure during loaded contact.

Deterministic repeats establish reproducibility for the tested inputs. They are not independent samples of real

hardware. A small variation campaign samples named cells of the frozen catalogue and reports that coverage, not an invented success probability over ten thousand tasks.

7 Recorded experimental results

7.1 Contact and ink time refinement

Same prescribed contact at 0.5 and 0.25 ms; actual deposited volume and erased fraction.
Recorded outcome: Passed.

Observable	Value
relative errors / deposited ink	0.01247183
relative errors / erased fraction	4.644939e-05

7.2 Ink-grid refinement

4096 versus 8192 output texels, unchanged physical stroke.
Recorded outcome: Passed.

Observable	Value
duration s	0.6
load n	1.2
velocity m s	0.08
relative error	0.01531326

7.3 Resolved material refinement

Independent nib, felt and cap specimens; actual saved FE solutions.
Recorded outcome: Passed.

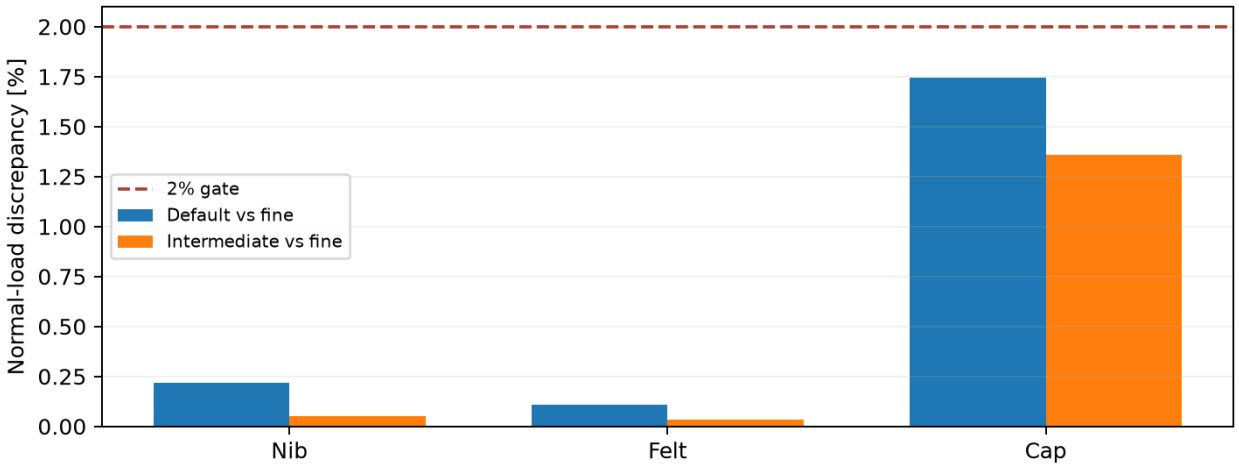


Figure 2: Measured material-mesh refinement. Each material has its own prescribed contact specimen; this is numerical convergence, not product calibration.

7.4 Pressure, speed and angle

All 27 prescribed writing conditions; each has time and space refinement.
Recorded outcome: Passed.

Observable	Value
wall s	85.35642

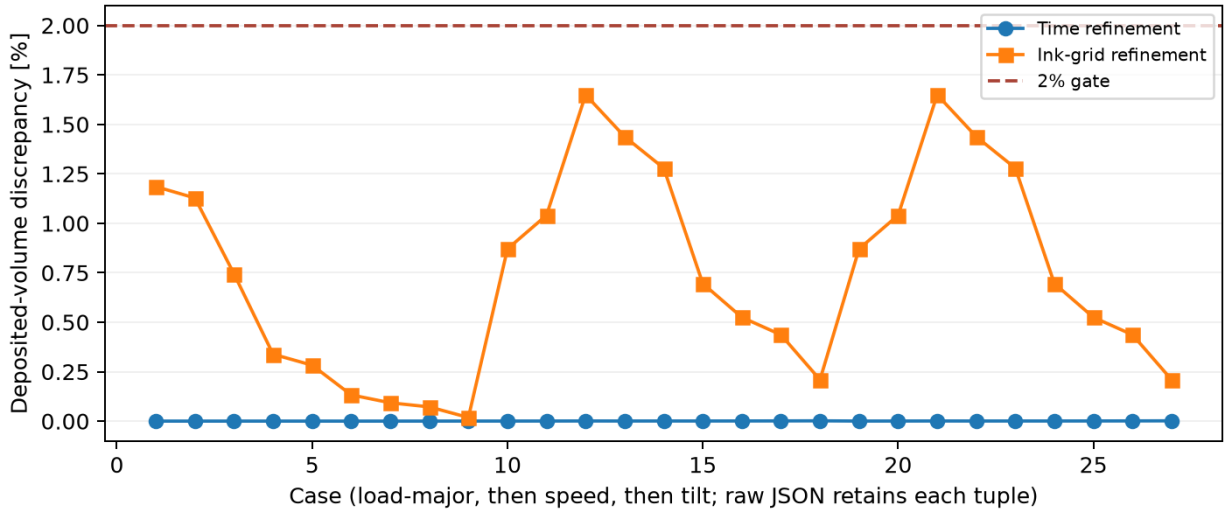


Figure 3: All 27 measured writing cases: 0.3/1.2/2 N, 0.02/0.08/0.2 m/s and 0/30/60 degree tilt. No failed case is omitted.

7.5 Loaded unwelded marker hold and release

Fixed-palm apparatus; the complete marker is retained by native articulated finger contact. Opening is followed by a free drop.
Recorded outcome: Passed.

Observable	Value
hold duration s	5
displacement m	0.00269755
maximum joint effort ratio	0.2543215
minimum contact distance during hold m	-9.71005e-05
writing load apparatus / axial n	1.2
writing load apparatus / tangential n	0.42
released drop m	0.4310288

7.6 Hand-held cap removal

Floating palm support and independent cap gauge; no tool support after setup.
Recorded outcome: Passed.

Observable	Value
ticks	5000
hand contact ticks	3999
peak slip m	0.001998016
cap release / tick	1215
cap release / pull n	2.14

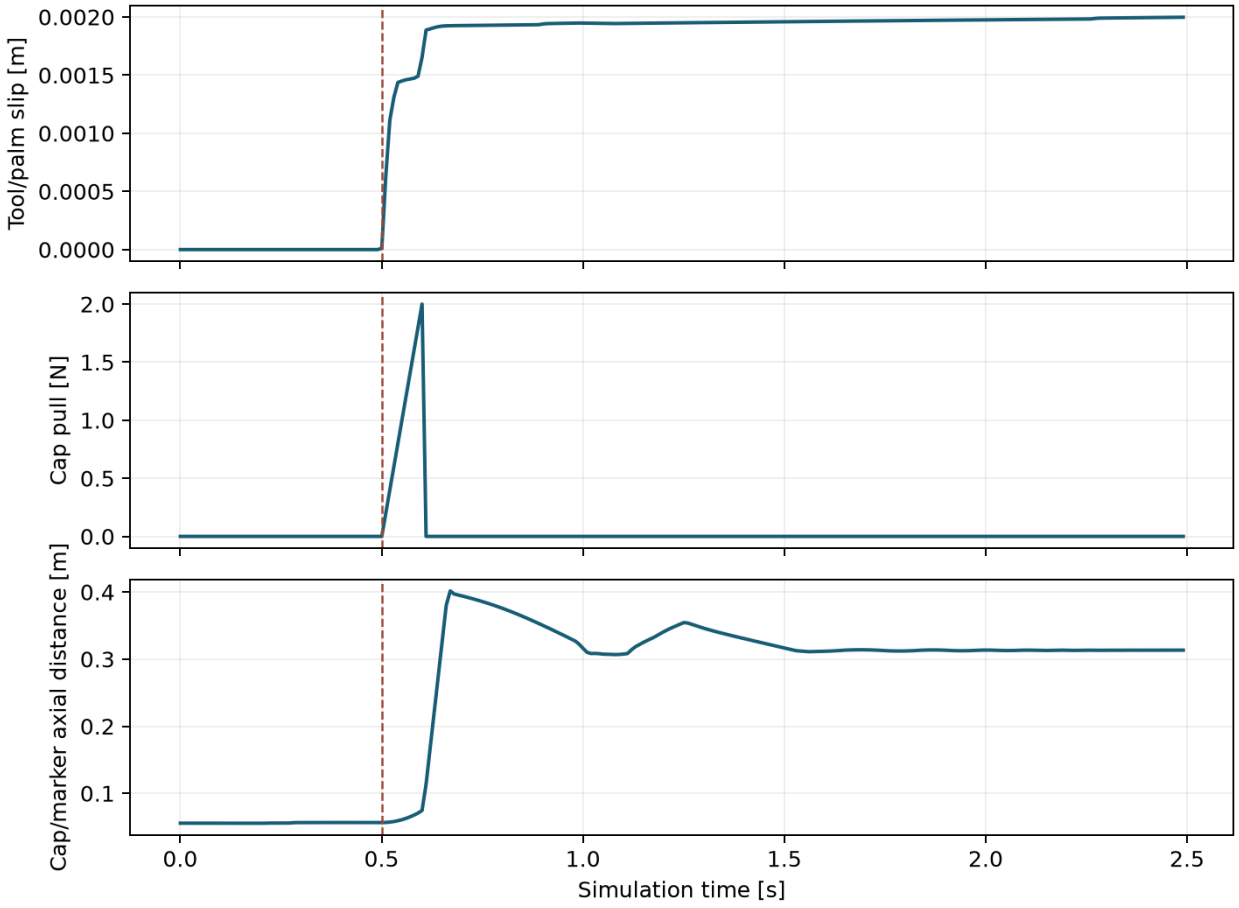


Figure 4: Hand-held cap removal. Recorded samples; dashed line marks removal of tool setup support. No interpolated physics is inserted.

7.7 Hand-held erasing

Known pigment is specimen preparation. Native fingers hold the eraser during the supported palm's motion. **Recorded outcome:** Passed.

Observable	Value
ticks	8500
board contact ticks	2627
peak board load n	1.92626
hand contact ticks	7500
peak slip m	0.006717828
seeded pigment ul	5.85339
removed ul	3.792649

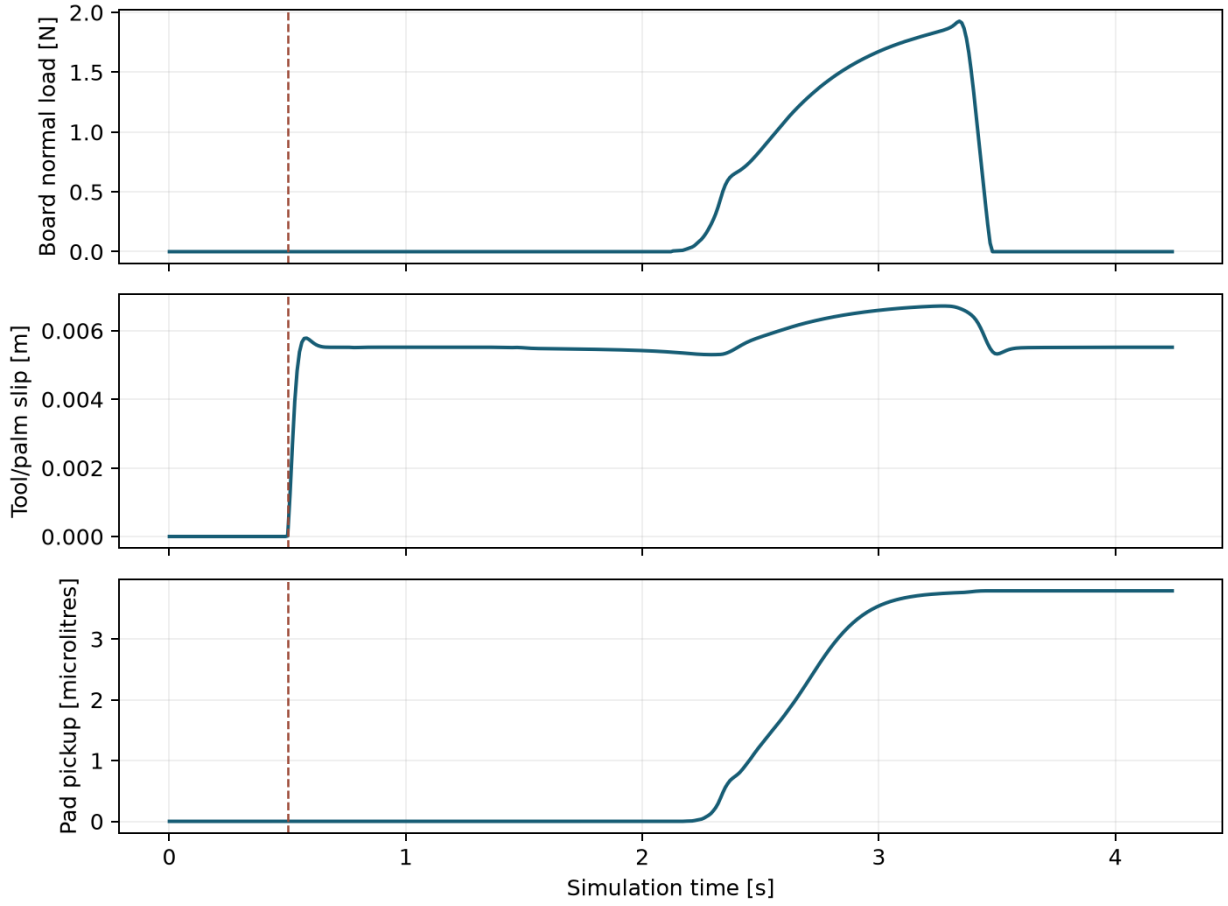


Figure 5: Hand-held erasing. Recorded samples; dashed line marks removal of tool setup support. No interpolated physics is inserted.

7.8 Complete-robot writing contact

Complete floating G1, recorded pelvis suspension and wrist load controller. This is not a robot task solution. **Recorded outcome:** Passed.

Observable	Value
ticks	5000
board contact ticks	1002
hand contact ticks	4000
peak slip m	0.002582639
peak board load n	0.6126065
ink ul	0.103342
peak joint effort ratio	1
contact step wall per sim s	84.89205
noncontact step wall per sim s	64.88352
wall s	250.0119

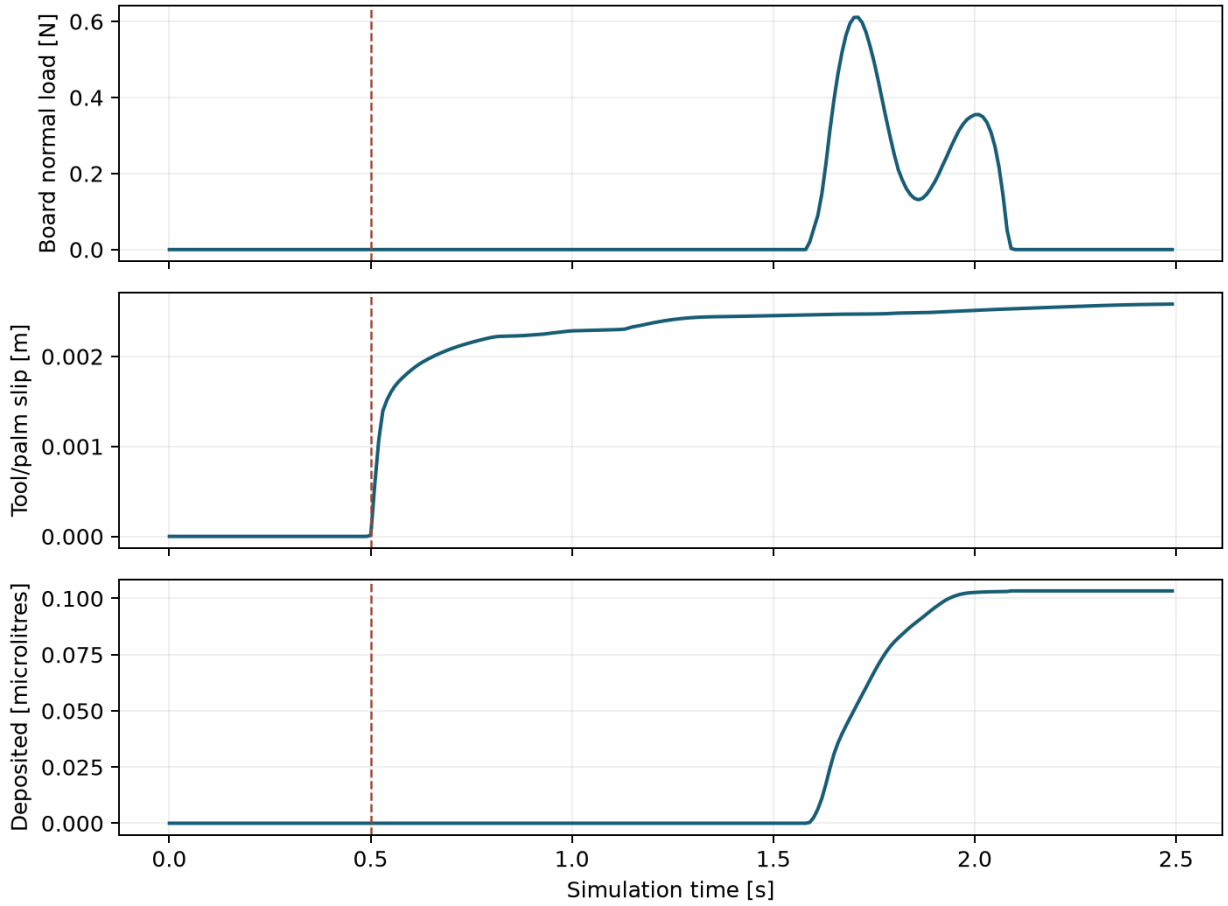


Figure 6: Complete-robot writing contact. Recorded samples; dashed line marks removal of tool setup support. No interpolated physics is inserted.

7.9 Free magnetic support and cap mechanics

Separate apparatus isolates magnetic support and cap extraction.
Recorded outcome: Passed.

Observable	Value
magnetic duster / duration s	3
magnetic duster / vertical drop m	0.0009977331
magnetic duster / normal gap m	-0.0005707461
cap release / applied pull n	2.33

7.10 Floating G1 under external wrist load

A separate board-equivalent force test, not actual nib contact.
Recorded outcome: Passed.

Observable	Value
whole robot welded	False
max linear impulse residual ns	0.0002138107
summed energy residual j	0.003158334

7.11 Independent rigid conservation

Independent rigid-body impulse/work checks. Friction integration is reported separately.
Recorded outcome: Passed.

Δt [ms]	Impulse residual [N s]	Work residual [J]	Energy drift [%]
0.5	1.07346e-07	6.28068e-06	0.0306562
0.25	6.07681e-08	2.0969e-06	0.0153258

7.12 Friction integration

Scalar counterexample using identical physical parameters, not an entire eraser trajectory.
Recorded outcome: Passed.

Observable	Value
dt s	0.0005
mass kg	0.03
normal load n	1.4
friction	0.55
explicit rest linear stability limit s	7.792208e-05
analytic equilibrium velocity m s	-0.0004136106

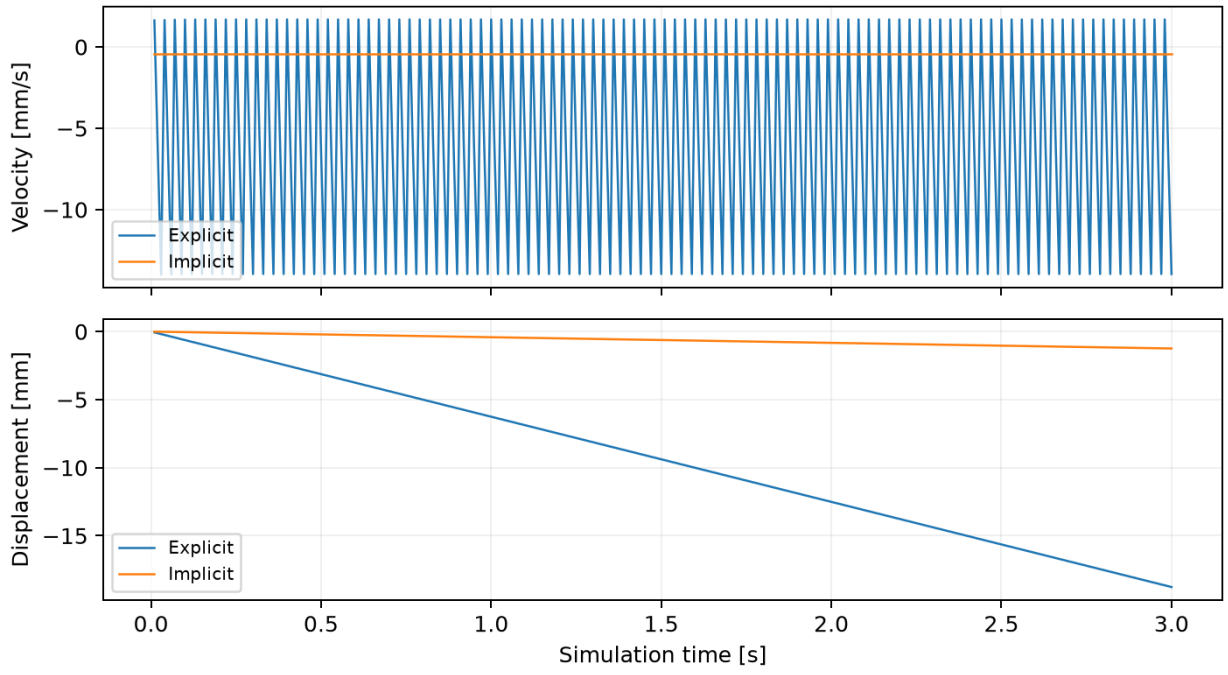


Figure 7: Scalar friction-integration counterexample under identical parameters. This is not a full eraser trajectory; the explicit method is shown as a rejected numerical reference.

7.13 Full HTTP, RGB and durable-history operating cost

Isolated measurement; four 1000-tick actions. Default full physics and every-tick durable recording remain enabled.
Recorded outcome: Passed.

Observable	Value
build s	21.6656
renderer startup s	84.55671
wall per sim s	90.41864
recording / failed trace	0
recording / eviction / evicted trace	0

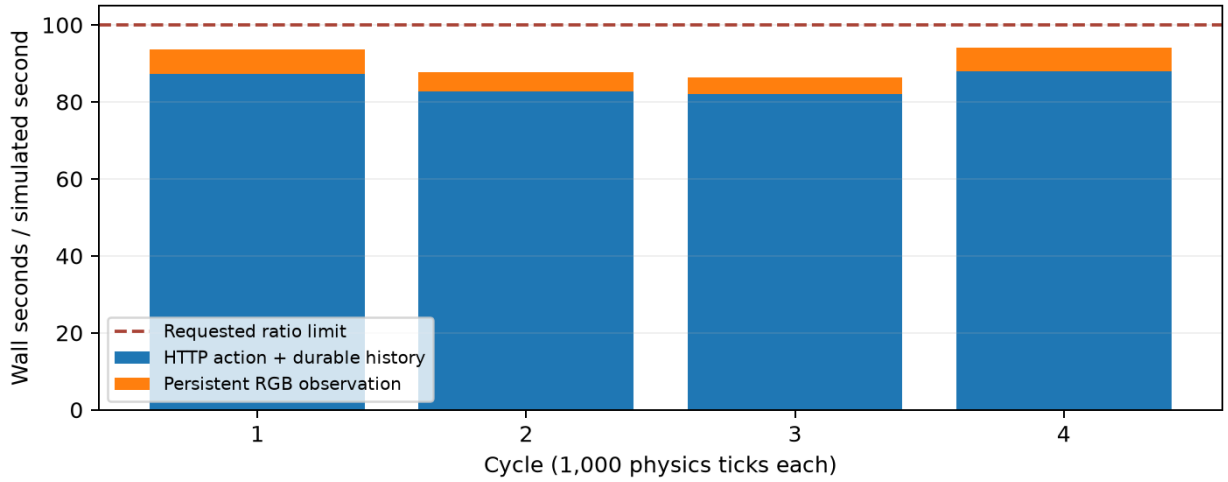


Figure 8: Measured full-scene operating cost. Renderer startup is excluded from these warm cycles and reported separately; every physics tick remains durably recorded.

7.14 Four-world CPU throughput

Exactly four complete concurrent CPU worlds with durable history; no RGB in this throughput measurement. Not a per-world latency claim.
Recorded outcome: Passed.

Observable	Value
requested worlds	4
actual worlds	4
slowest world wall s	53.172
aggregate simulated s	2
aggregate wall per sim s	26.586

World	Wall time [s]	Simulated time [s]	Per-world ratio
0	53.0328	0.5	106.066
1	52.0643	0.5	104.129
2	53.172	0.5	106.344
3	52.9769	0.5	105.954

7.15 GPU execution and replay

Full scene with explicit backend settings. GPU support does not imply a speedup.
Recorded outcome: Passed.

Observable	Value
cpu / step wall s	3.884
cuda:0 / step wall s	527.4259
comparison / max position error m	8.735217e-05
cpu / replay repetitions	5
cuda:0 / replay repetitions	5

7.16 Exact four-world native GPU batch

A distinct native batched experiment, not CPU-process sharding.
Recorded outcome: Passed.

Observable	Value
summary / actual batch	4
summary / warm ticks each	100
summary / warm wall s	34.43992
summary / world steps per wall s	11.61443

7.17 Persistent physical scene

One continuous 60000-tick passive scene audit; every tick checked.
Recorded outcome: Passed.

Observable	Value
duration s	30
physics ticks	60000
continuous	True
check stride ticks	1

7.18 Variation coverage

Twelve predetermined representatives, not a claim about all 10000 tasks.
Recorded outcome: Passed.

Observable	Value
count	12

7.19 Crash recovery and exact continuation

Separate-process abrupt termination and durable recovery, not a continuous video.
Recorded outcome: Passed.

Observable	Value
future bit exact	True

7.20 Actual HTTP cameras and role separation

Real loopback requests, visible ink updates, reset and role filtering.
Recorded outcome: Passed.

Observable	Value
written red pixels	1438
restored red pixels	0
robot rgb mean absolute change	4.460709

7.21 Manual interface and integration regression

Software contract checks accompany, but do not replace, the physical experiments above. Real browser results are separately retained.
Recorded outcome: Passed.

Observable	Value
tests run	103
failures	None
errors	None
skipped	None

8 Rendered evidence



Figure 9: Default classroom, actual RTX render. Review camera only; no scene rearrangement for a task solution.

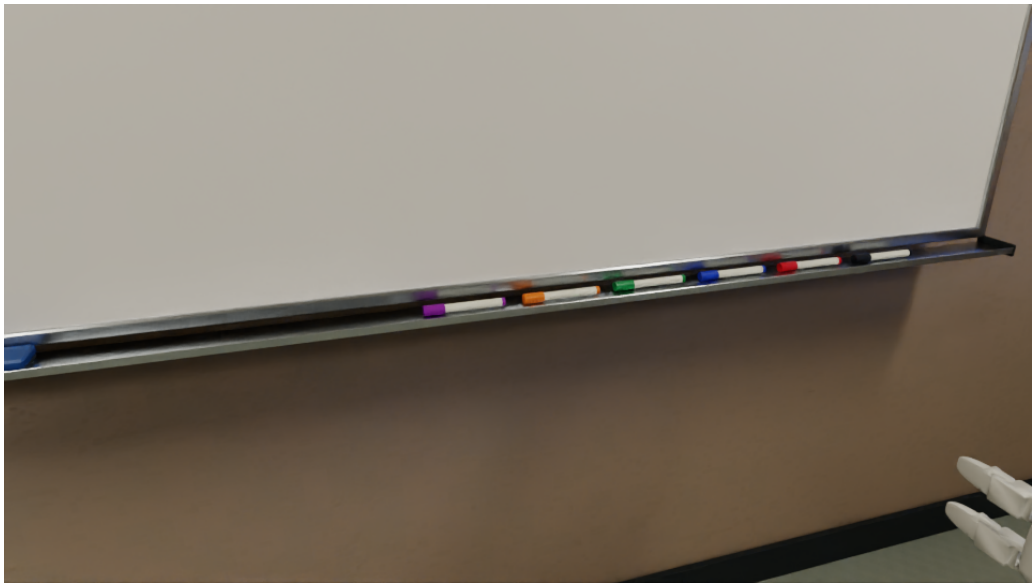


Figure 10: Tool dimensions, printed barrel finish, tray support and contact shadows in the actual renderer.

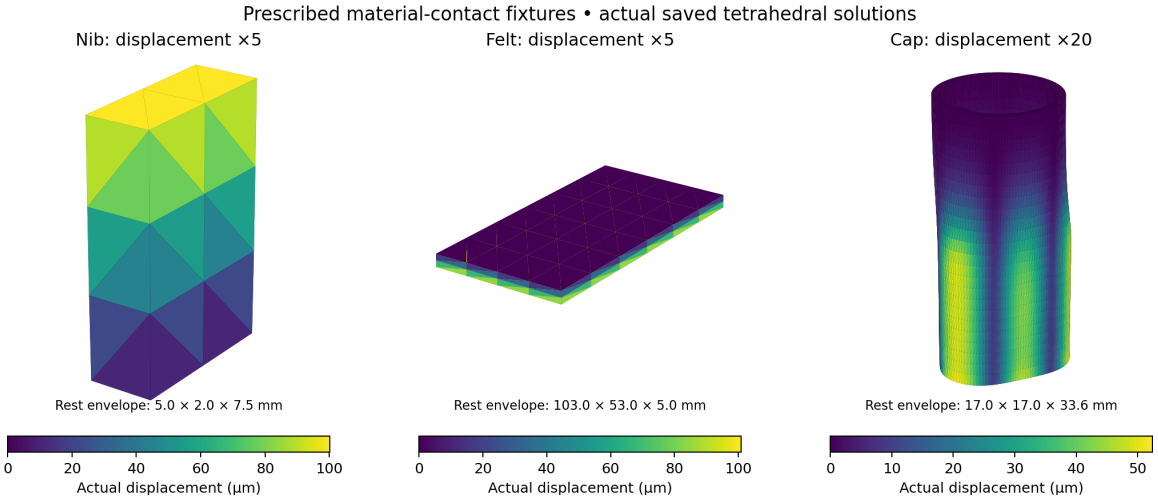


Figure 11: Saved material solutions. Displacement magnification is labelled; the colour scale gives actual displacement.



Figure 12: Actual physically deposited pigment uploaded to the live board material, not an image overlay.

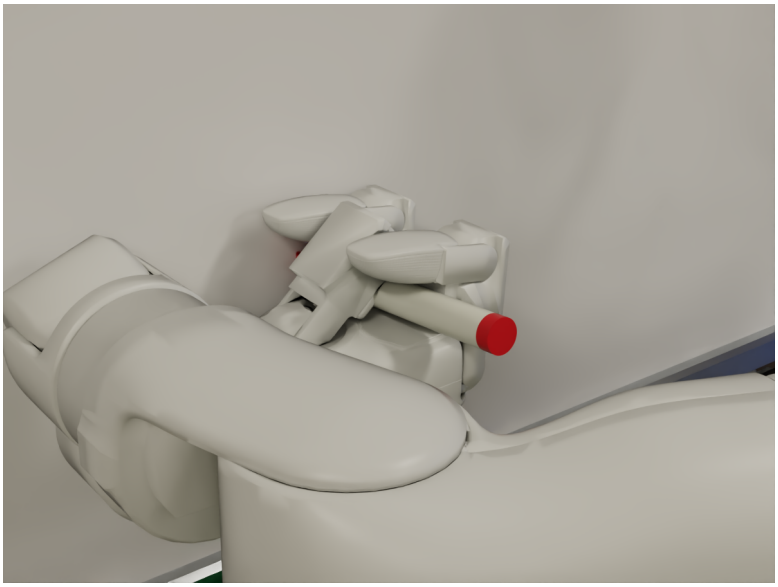


Figure 13: Complete G1 contact apparatus at tick 4000: native fingers, uncapped marker and board.

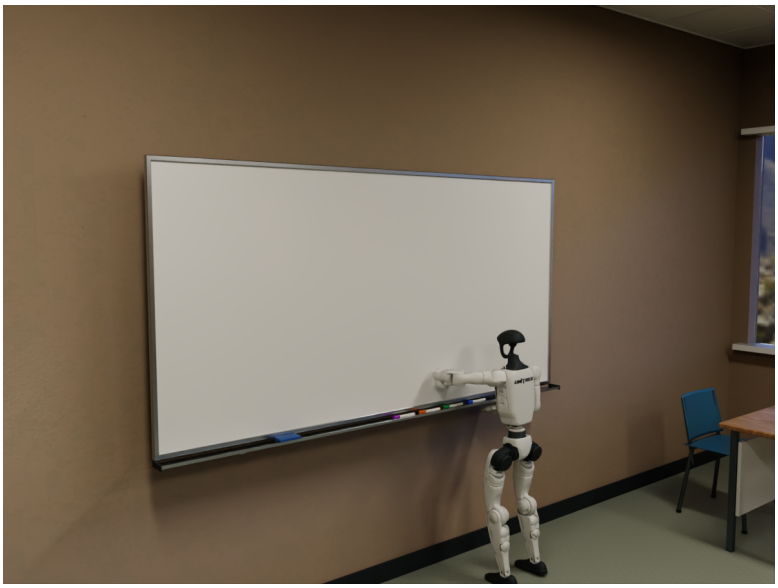


Figure 14: Wider view of the same uninterrupted complete-robot component experiment.

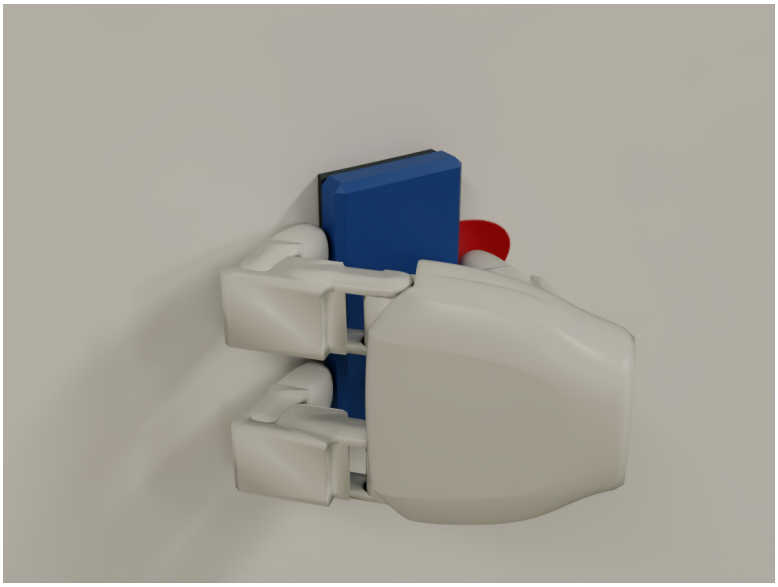


Figure 15: Actual articulated-hand erasing experiment; seeded pigment and recorded palm support are disclosed apparatus.

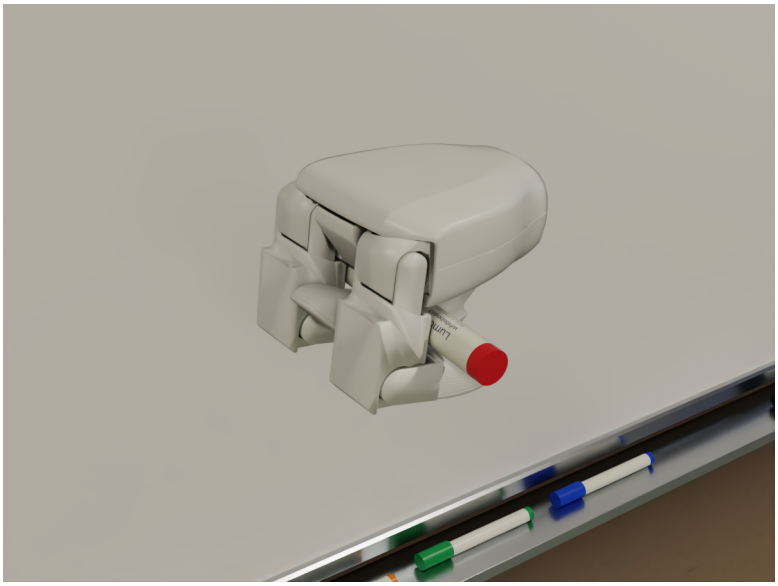


Figure 16: Cap extraction fixture after the independent pull gauge removed the cap; no marker attachment.



Figure 17: Loaded retention/release experiment: one of 580 real recorded frames, not a restored state.

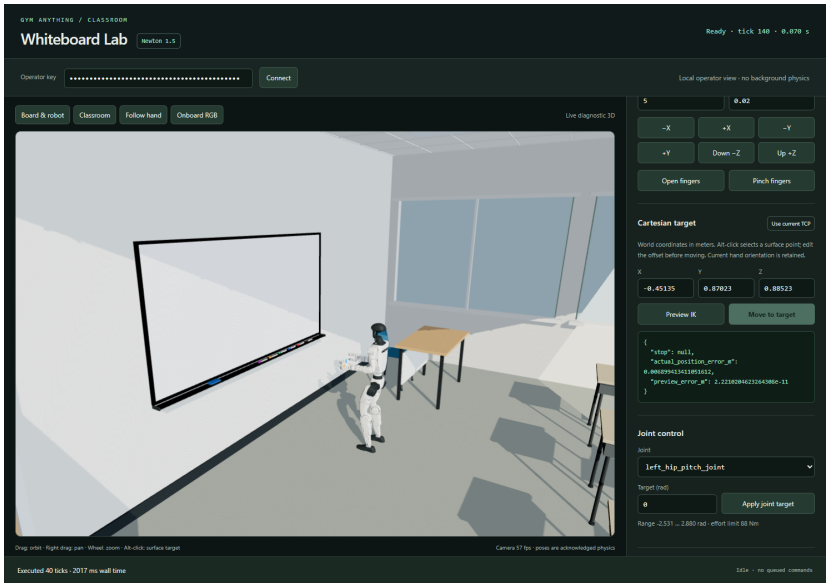


Figure 18: Actual isolated-browser interaction evidence; diagnostic UI, not an onboard policy camera.

9 Visual evidence and remaining model boundaries

Material and lighting work must be judged in actual rendered images. The room uses three-dimensional furniture, frame bevels, mounted light panels, daylight through physical windows, a glass material with finite thickness, and distinct coated board, polymer, felt, metal and wood responses. CC0 appearance maps have asset-specific attribution, source hashes and physical tile scales[10]. Reflectance models organize these choices; they do not turn guessed roughness into a measured property[11].

Review images include a wide classroom, tools at working distance, close-up finger contact, the evolving stroke or erased region and the onboard camera. Inspection cameras are labelled privileged. Camera repositioning is not a fix for the robot's task: the onboard mount stays tied to the selected hardware. The manual WebGL inspection view and offline RTX renders have different shading quality, but must agree on object transforms and the visible ink state.

The package has deliberate physical limits. The board and wall are fixed boundaries, not flexing structures. Nib and pad deformation is local, quasi-static small-strain elasticity, not full porous-material dynamics. Rigid external proxies still represent some tool contacts. Drying is a reduced surface-kinetic model; cap-off dry-out, solvent transport, felt wear, pigment redeposition and permanent stains are absent. The sensor is idealized RGB, not a full electronics and optical-noise simulator. Numerical strain or convergence failures must be reported as failed state, with executed-tick and history information; they are not valid task outcomes. Transfer uses a prescribed projected tool footprint centred on the load-weighted contact location, rather than deriving the inked area from individual loaded fibres or tetrahedral contact nodes. In particular, partial edge contact is not a resolved microscopic contact-area model. The pressure/speed/angle audit checks this implemented transfer law, not a stronger claim about physical ink chemistry.

These boundaries do not remove the requirement to implement and test all claimed features. A missing working snapshot, a stale camera image, duplicated contact response or an incomplete report is a package defect, not an unavoidable consequence of digital simulation. Conversely, absence of a physical laboratory or an autonomous solving policy is not a reason to leave the package unfinished.

10 Modules and reproduction

10.1 Module map

Modules	Responsibility
`config.py`, asset lock, robot import	SI configuration, exact versions, named-product dimensions, G1 assembly and limits.
`env.py`, solver and contact modules	One stepping authority, contact ownership, finite-state and execution guards.
`elasticity.py`, `deformable_contact.py`, `friction.py`	Material stiffness, unilateral contact, friction and rigid-body wrench coupling.
`ink.py`, `tasks.py`	Finite pigment state, transfer, wiping and independent target scoring.
`sensors.py`, `usd_scene.py`, renderer modules	Hardware-relative optical pose, shared scene export, current-state RGB without physics.
`api.py`, GAR backend, dashboard	Consistent action/observation semantics and role-filtered interactive access.
`history.py`, `storage.py`, trace codec	Bounded per-tick data, durable commands, checkpoints and explicit incomplete history.

10.2 Reproduction

The release pins Newton 1.5.0 and the robot assets. Renderer execution uses its own Isaac Sim 5.1 interpreter. Start the environment from the release with `gar run whiteboard --port 8080`. The component experiments reported here ran independently and serially; the campaign is not represented as one continuous robot solution.

Each experiment records its effective configuration and raw measurements. The summary numbers and plots in this report are generated from those records, and failed required experiments stop the report build.

References

[1] MooreCo, *Porcelain Steel Whiteboards, Deluxe Aluminum Trim, 202AX*, June 2024 specification. https://www.moorecoinc.com/wp-content/uploads/2024/06/Porcelain-Steel-Whiteboards-Deluxe-Aluminum-Trim-202AX-Spec-Sheet_551927c3061.pdf.

[2] STAEDTLER Japan, *Lumocolor whiteboard markers*, 351 B individual-product specification. https://www.staedtler.jp/products/pen02/whiteboard.html. Rechecked 9 September 2026.

[3] STAEDTLER, *Lumocolor whiteboard wiper 652*. https://www.staedtler.com/au/en/products/markers/whiteboard-markers/lumocolor-whiteboard-wiper-652-whiteboard-wiper-652-bk/. Rechecked 9 September 2026.

[4] Unitree Robotics, *G1 specifications*. https://www.unitree.com/g1/. Configuration-dependent published limits; rechecked 9 September 2026.

[5] Unitree Robotics, *G1 revision-specific robot descriptions*. https://github.com/unitreerobotics/unitree_ros/tree/master/robots/g1_description. Local URDF and mesh revisions are fixed by the package asset manifest.

[6] RealSense, *D435 camera*. https://www.realsenseai.com/products/stereo-depth-camera-d435/. RGB and depth specifications are distinct; rechecked 9 September 2026.

[7] R. Liu, W. Wan, K. Koyama and K. Harada, *Multi-Pen Robust Robotic 3D Drawing Using Closed-Loop Planning*, 2020. https://arxiv.org/abs/2009.14501.

[8] D. Song, T. Lee and Y. J. Kim, *Artistic Pen Drawing on an Arbitrary Surface Using an Impedance-Controlled Robot*, ICRA 2018. https://daeunsong.github.io/publications/paper/icra18.pdf.

[9] X. Guo and colleagues, *Flying Calligrapher: Contact-Aware Motion and Force Planning and Control for Aerial Manipulation*. https://xiaofeng-guo.github.io/flying-calligrapher/.

[10] Poly Haven, CC0 asset licence and asset metadata. https://polyhaven.com/license. Individual assets, authors and hashes are listed in `visual_assets.py` and the third-party notice.

[11] B. Burley, *Physically-Based Shading at Disney*, SIGGRAPH course notes, 2012. https://disneyanimation.com/publications/physically-based-shading-at-disney/.

Further sources. The research behind this model drew on 31 sources, including manufacturer classroom-furniture references, ink/material disclosures and versioned rendering documentation. Product photographs guide construction but are not presented as metrology, and third-party photographs are not redistributed as scene textures.